Game Theory in Electricity Markets

Exercise Sheet 2 — Nash Equilibria

11th of September 2026

Exercise 1 Hawk and Dove 2

Please read carefully, the game setting has changed slightly to last week.
Two birds contest a resource of value $V > 0$ (e.g., food, territory). Each animal simultaneously chooses to act aggressively (**Hawk**, H) or passively (**Dove**, D).

- If both play *Dove*, they share the resource peacefully: each gets $V/2$.
- If one plays *Hawk* and the other *Dove*: the Hawk takes the whole resource, the Dove retreats and gets nothing.
- If both play *Hawk*, they fight over it, until they realize they are equally matched, share and both get half of the resource. The catch: fighting comes not for free, it costs both of them C. So in total they both get a payoff $V/2 - C$.

Throughout this exercise, assume that fights are costly, i.e. $C > V$.

		Player 2	
		H	D
Player 1	H	$\left(\frac{V}{2} - C,\ \frac{V}{2} - C\right)$	$(V,\ 0)$
	D	$(0,\ V)$	$\left(\frac{V}{2},\ \frac{V}{2}\right)$

Task: Calculate the mixed Nash equilibria of this game.

Exercise 2 Bertrand Competition

Two firms, 1 and 2, compete to serve a single, unit of demand from a buyer whose valuation of the good is 1. Each firm simultaneously and independently sets a price $p_i \in [0, 1]$. Production is costless for both firms. The buyer purchases the unit from whichever firm charges the lower price; if $p_1 = p_2$, the buyer's purchase (and hence the unit) is split equally between the two firms. A firm's payoff is its revenue: price times quantity sold.

(2.1) Find all pure Nash equilibria of this game.

Now assume both firms additionally face a **capacity constraint**: each firm can supply at most $\frac{2}{3}$ of a unit. The rules of demand allocation change accordingly:

- If $p_i < p_j$, firm i sells at its full capacity, $\frac{2}{3}$; the remaining demand, $\frac{1}{3}$, is then served by firm j at price p_j (note that $\frac{1}{3}$ does not exceed firm j's capacity, so it is served in full).
- If $p_1 = p_2$, demand is split equally as before: each firm sells $\frac{1}{2}$ unit (which is within its capacity).

(2.2) Calculate a mixed Nash equilibrium of this game. *Hint: the game is symmetric.*

Exercise 3 Electricity Market Bidding

Recall the electricity market from the slides of week 1, in which generators submit bids into a uniform-price (merit-order) auction.

There are five generators, G_i, each with a marginal cost c_i (in EUR/MWh) and a capacity q_i:

	G_1	G_2	G_3	G_4	G_5
Marginal cost c_i	5	12.5	25	40	65
Capacity q_i	100	40	50	30	30

Total demand is fixed at $D = 180$ and must be served. Each generator i simultaneously submits an integer bid $b_i \in \{0, 1, 2, \ldots, 80\}$ (EUR/MWh). The market is cleared as in the lecture:

- **Merit order:** generators are ranked from the lowest to the highest bid, and dispatched in that order until the demand is covered.
- **Ties:** if two (or more) generators submit the same bid and only part of their combined capacity is needed to meet the remaining demand, that remaining demand is split *equally* between the tied generators.
- **Price:** every dispatched generator is paid the same price, equal to the bid of the last (marginal) generator needed to meet demand.
- **Profit:** generator i's profit is $\pi_i(b) = (\text{price} - c_i) \times q_i^{\text{dispatched}}$, where $q_i^{\text{dispatched}}$ is the quantity dispatched to i. A generator that is not dispatched at all earns 0.

Task: Implement the *best-response algorithm* to find a pure Nash equilibrium of this bidding game.

Task: Think about how to calculate the price of anarchy of a mixed Nash equilibrium and calculate it if you found one.

Note: Try to not hardcode any of the values such that you can consider different market settings.

Note: A very barbaric way to check if a game has multiple Nash equilibria is to just start best-response from multiple points. Try that.

Exercise 4 For next week

Read up on the following topics:

- Correlated equilibria: what are they, do they always exist,
- Coarse correlated equilibria: what are they, do they always exist.

Try to understand what the main distinction between those classes and mixed Nash equilibria is.